Quantum Harmonic oscillator

$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$

$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{q}^2$ $\qquad [\hat{q}, \hat{p}] = \hat{q}\hat{p} - \hat{p}\hat{q} = i\hbar$

$\hat{a} = \frac{1}{\sqrt{2m\hbar\omega}} (m\omega \hat{q} + i\hat{p})$

$\hat{a}^+ = \frac{1}{\sqrt{2m\hbar\omega}} (m\omega \hat{q} - i\hat{p})$

Non Hermitian adimensional operators

$\hat{a}^+$ is the adjoint of $\hat{a}$

$[\hat{a}\, \hat{a}^+] = \hat{a}\hat{a}^+ - \hat{a}^+\hat{a} = 1$

$\hat{H} = \frac{1}{2}\hbar\omega(\hat{a}^+\hat{a} + \hat{a}\hat{a}^+) = \underline{\hbar\omega(\hat{a}^+\hat{a} + \frac{1}{2})}$ → Expression of $\hat{H}$ in terms of $\hat{a}$ & $\hat{a}^+$

$|n\rangle$ is an eigenstate of $\hat{H}$ with eigenvalue E_n

$\hat{H}|n\rangle = E_n|n\rangle$

$\hat{H}\hat{a}^+|n\rangle = \dots = (E_n + \hbar\omega)\hat{a}^+|n\rangle \Rightarrow$ $\hat{a}^+|n\rangle$ is an eigenstate of $\hat{H}$ with eigenvalue increased by $\hbar\omega \Rightarrow E_n \to E_n + \hbar\omega$

$\hat{H}\hat{a}|n\rangle = \dots = (E_n - \hbar\omega)\hat{a}|n\rangle \Rightarrow$ $\hat{a}|n\rangle$ is an eigenstate of H with eigenv. $E_n - \hbar\omega$

Assume that $|0\rangle$ is the fundamental level $\Leftrightarrow$ $\hat{H}|0\rangle = E_0|0\rangle$

$\hat{a}|0\rangle = 0$ (Null) $\Rightarrow \hbar\omega(\hat{a}^+\hat{a} + \frac{1}{2})|0\rangle = E_0|0\rangle$ $\quad \cancel{\hbar\omega\hat{a}^+\hat{a}|0\rangle} + \frac{1}{2}\hbar\omega|0\rangle = E_0|0\rangle \Rightarrow E_0 = \frac{1}{2}\hbar\omega$

<u>Normalization</u> requires that:

$\hat{a}^+|n\rangle = \sqrt{n+1}\,|n+1\rangle$ ↑ $\Rightarrow \langle n+1|\hat{a}^+|n\rangle = \sqrt{n+1}$

$\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle$ ↓ $\qquad \langle n-1|\hat{a}|n\rangle = \sqrt{n}$

Only non vanishing matrix elements of $\hat{a}$ & $\hat{a}^+$

<u>Number operator</u>

$\hat{H}|n\rangle = \hbar\omega(\hat{a}^+\hat{a} + \frac{1}{2})|n\rangle = E_n|n\rangle = \hbar\omega(n + \frac{1}{2})|n\rangle$ let's define the operator $\hat{n} = \hat{a}^+\hat{a}$

↓

<u>number operator</u>

$\hbar\omega(\hat{n} + \frac{1}{2})|n\rangle = \hbar\omega(n + \frac{1}{2})|n\rangle \Rightarrow \hat{n}|n\rangle = n|n\rangle$

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Quadrature operators

$$\hat{X} = \sqrt{\frac{m\omega}{2\hbar}}\,\hat{q} = \tfrac{1}{2}(\hat{a}^+ + \hat{a})$$
$$\hat{Y} = \frac{1}{\sqrt{2m\hbar\omega}}\,\hat{p} = \frac{i}{2}(\hat{a}^+ - \hat{a})$$

Hermitian operators that express $\hat{q}$ and $\hat{p}$ in adimensional form

$$\Rightarrow \begin{cases} \hat{a}^+ = \hat{X} - i\hat{Y} \\ \hat{a} = X + i\hat{Y} \end{cases}$$

$\hat{X}$ and $\hat{Y}$ do not commute: $[\hat{X},\hat{Y}] = \frac{1}{2\hbar}[\hat{q},\hat{p}] = \frac{1}{2\hbar}\cdot i\hbar = i/2$

$$\hat{H} = \hbar\omega\left(\hat{a}^+\hat{a} + \tfrac{1}{2}\right) = \hbar\omega\left[(\hat{X}-i\hat{Y})(\hat{X}+i\hat{Y}) + \tfrac{1}{2}\right] = \hbar\omega\left[\hat{X}^2 + \hat{Y}^2 + i(\hat{X}\hat{Y} - \hat{Y}\hat{X}) + \tfrac{1}{2}\right] = \hbar\omega(\hat{X}^2 + \hat{Y}^2)$$

$\hat{X}$ and $\hat{Y}$ allow us to express the uncertainty Einsteiner principle in adimensional form

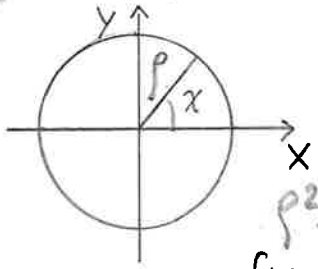
$$(\Delta q)^2(\Delta p)^2 \geq \hbar^2/4 \iff (\Delta X)^2(\Delta Y)^2 \geq 1/16$$

Variance $(\Delta f)^2 = \langle f^2\rangle - \langle f\rangle^2$

$$(\Delta f)^2(\Delta g)^2 \geq \left\{\tfrac{1}{2}\langle[\hat{F},\hat{G}]\rangle\right\}^2$$

Pictoral representation of Energy

$\hat{H} = \hbar\omega(\hat{X}^2 + \hat{Y}^2)$ $\quad \hat{H}|n\rangle = E_n|n\rangle$



$\rho^2 = x^2 + y^2$

$$\begin{cases} x = \rho\cos\chi \\ y = \rho\sin\chi \end{cases}$$

$\hat{H}/\hbar\omega = \hat{X}^2 + \hat{Y}^2$ $\quad \rho = \sqrt{\frac{E_n}{\hbar\omega}}$

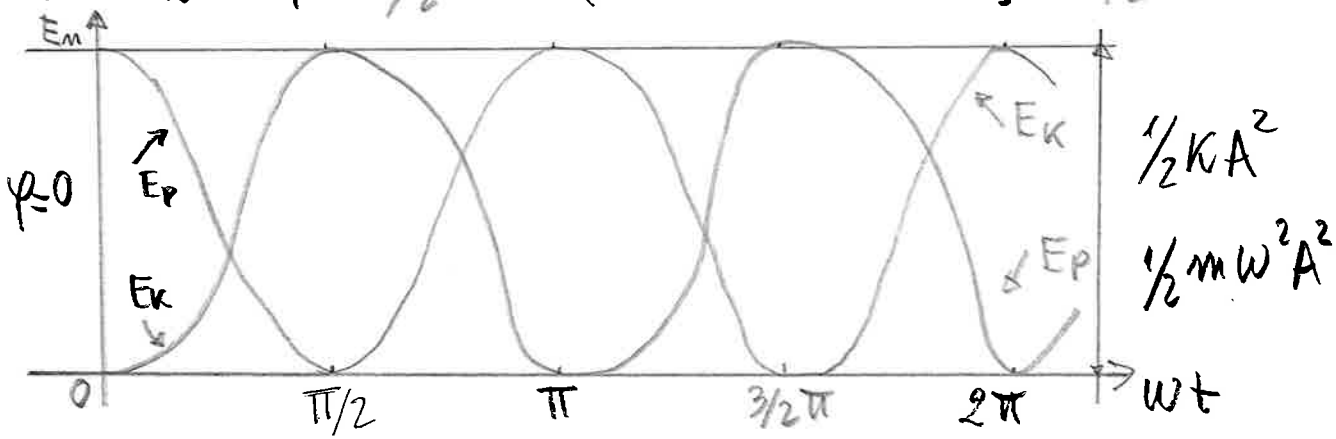
$$\begin{cases} x^2 = E_n/\hbar\omega\cdot\cos^2\chi \\ y^2 = E_n/\hbar\omega\cdot\sin^2\chi \end{cases}$$

! Not correct since χ cannot be determined

Classic harmonic oscillator

$x(t) = A\sin(\omega t + \varphi)$ $\quad \omega = \sqrt{\frac{K}{m}}$

$$E_n = E_K + E_p = \tfrac{1}{2}KA^2(\sin^2\omega t + \cos^2\omega t) = \tfrac{1}{2}KA^2$$



Kinetic an potential energies are out of phase by $\pi/2$

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EM Field in a multimode (quantization) cavity. Quantum repres.

- Each mode is a quantum oscillator independent from one another
- Using the formalism of operators $\hat{a}$ & $\hat{a}^+$ we establish the correspondence

Mode $\vec{k},\lambda \rightarrow \begin{cases} A_{k\lambda} \rightarrow \hat{a}_{k\lambda} & A_{k\lambda}, A^*_{k\lambda} \text{ complex amplitude of mode vector potential} \\ A^*_{k\lambda} \rightarrow \hat{a}^+_{k\lambda} & \hat{a}_{k\lambda}\ \hat{a}^+_{k\lambda} \text{ destruction \& creation operators} \end{cases}$

Hamiltonian of mode k,λ — Classical energy of mode $k\lambda$

$$\hat{H}_{k\lambda} = \tfrac{1}{2}\hbar\omega\left(\hat{a}_{k\lambda}\hat{a}^+_{k\lambda} + \hat{a}^+_{k\lambda}\hat{a}_{k\lambda}\right) \Longleftrightarrow \mathcal{E}_{k\lambda} = \varepsilon_0 V \omega_k^2 \left(A_{k\lambda}A^*_{k\lambda} + A^*_{k\lambda}A_{k\lambda}\right)$$

Energy conservation requires that:

$$A_{k\lambda} \rightarrow \sqrt{\frac{\hbar}{2\varepsilon_0 V \omega_k}}\,\hat{a}_{k\lambda} \qquad A^*_{k\lambda} \rightarrow \sqrt{\frac{\hbar}{2\varepsilon_0 V \omega_k}}\,\hat{a}^+_{k\lambda}$$

Eigenstates and eigenvalues of the Hamiltonian of a single mode $k\lambda$

$\hat{H}_{k\lambda}|n_{k\lambda}\rangle = E_{n_{k\lambda}}|n_{k\lambda}\rangle$ $\quad |n_{k\lambda}\rangle$ eigenstate of H, $E_{n_{k\lambda}} = \hbar\omega\left(n_{k\lambda} + \tfrac{1}{2}\right)$

energy in the mode $k\lambda$

Number operator & operators $\hat{a}, \hat{a}^+$

$\hat{n}_{k\lambda} = \hat{a}^+_{k\lambda}\hat{a}_{k\lambda}$ $\quad \hat{n}_{k\lambda}|n_{k\lambda}\rangle = n_{k\lambda}|n_{k\lambda}\rangle$ $\quad n_{k\lambda}$ = number of photon in $k\lambda$

$\hat{a}_{k\lambda}|n_{k\lambda}\rangle = \sqrt{n}\,|n_{k\lambda}-1\rangle$ destruction operator: removes 1 photon

$\hat{a}^+_{k\lambda}|n_{k\lambda}\rangle = \sqrt{n+1}\,|n_{k\lambda}+1\rangle$ creation operator: creates 1 photon

$|n_{k\lambda}\rangle$ single mode states called Number States or Fock states

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Quantum representation of the EM field in the cavity

Multimode state that can be factorized in the product of single mode states

$$|n_{k_1,\lambda_1}, n_{k_1,\lambda_2}, n_{k_2,\lambda_1}, \dots\rangle = |n_{k_1,\lambda_1}\rangle |n_{k_1,\lambda_2}\rangle |n_{k_2,\lambda_1}\rangle \dots = |\{n_{k\lambda}\}\rangle \quad n_{k\lambda}=0,1,2\dots \ \forall k,\lambda$$

$|\{n_{k\lambda}\}\rangle$ Multimode number states form a <u>complete set</u> of states for the EM field in the cavity when every member of the set goes from $0 \to \infty$ photons

$$|\text{Pure state}_{EM}\rangle = \sum_{\{n_{k\lambda}\}} C(\{n_{k\lambda}\}) |\{n_{k\lambda}\}\rangle \quad \leftarrow \text{Superposition of base states}$$

Expression of the fields $\vec{A}$, $\vec{E}_T$ & $\vec{B}$ in quantum formalism

$$\vec{A}(\vec{r},t) = \sum_k \sum_\lambda \vec{e}_{k\lambda} A_{k\lambda}(\vec{r},t)$$

$$A_{k\lambda}(\vec{r},t) = A_{k\lambda} \exp(-i\omega_k t + \vec{k}\cdot\vec{r}) + c.c.$$

$$\vec{E}_T(\vec{r},t) = \sum_k \sum_\lambda \vec{e}_{k\lambda} E_{k\lambda}(\vec{r},t)$$

$$E_{k\lambda}(\vec{r},t) = i\omega_k \left[A_{k\lambda} \exp(-i\omega_k t + \vec{k}\cdot\vec{r}) + c.c.\right]$$

$$\vec{B}(\vec{r},t) = \sum_k \sum_\lambda \frac{\vec{k}\times\vec{e}_{k\lambda}}{|\vec{k}|} B_{k\lambda}(\vec{r},t)$$

$$B_{k\lambda}(\vec{r},t) = ik\left[A_{k\lambda} \exp(-i\omega_k t + \vec{k}\cdot\vec{r}) + c.c.\right]$$

$$\hat{\vec{A}}(\vec{r},t) = \sum_{k,\lambda} \vec{e}_{k\lambda} \hat{A}_{k\lambda}(\vec{r},t)$$ Heisenberg picture operators depend on time

$$\hat{A}_{k\lambda}(\vec{r},t) = \sqrt{\frac{\hbar}{2\varepsilon_0 \omega_k V}} \left\{\hat{a}_{k\lambda} \exp[-i(\omega_k t - \vec{k}\cdot\vec{r})] + \hat{a}^+_{k\lambda} \exp[i(\omega_k t - \vec{k}\cdot\vec{r})]\right\}$$

$$\hat{E}_T(\vec{r},t) = \hat{E}^+_T(\vec{r},t) + \hat{E}^-_T(\vec{r},t) \qquad \chi_k = [\omega_k t - \vec{k}\cdot\vec{r} - \pi/2]$$

$$\hat{E}^+_T(\vec{r},t) = \sum_{k\lambda} \sqrt{\frac{\hbar\omega_k}{2\varepsilon_0 V}}\, \vec{e}_{k,\lambda}\, \hat{a}_{k\lambda} \exp[-i\chi_k(\vec{r},t)] \quad \oplus$$

$$\hat{E}^-_T(\vec{r},t) = \sum_{k\lambda} \sqrt{\frac{\hbar\omega_k}{2\varepsilon_0 V}}\, \vec{e}_{k,\lambda}\, \hat{a}^+_{k\lambda} \exp[+i\chi_k(\vec{r},t)] \quad \ominus$$

$$\hat{E}_T(\vec{r},t) = \sum_{k\lambda} \sqrt{\frac{2\hbar\omega_k}{\varepsilon_0 V}}\, \vec{e}_{k\lambda} \left\{\hat{X}_{k\lambda} \cos[\chi(\vec{r},t)] + \hat{Y}_{k\lambda} \sin[\chi(\vec{r},t)]\right\}$$

$$\hat{B}(\vec{r},t) = \hat{B}^+(\vec{r},t) + \hat{B}^-(\vec{r},t)$$

$$\hat{B}^+(\vec{r},t) = \dots \qquad \hat{B}^-(\vec{r},t) = \dots$$

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